

NIFTY100 FINANCIAL INTELLIGENCE

Analyst Guide

Sprint 6 — Clustering, REST API, Dashboard, QA and Sign-Off

Coverage: 92 companies • 11 peer groups • 1,188 financial-ratio rows • 5 analytical clusters • 92 company tearsheets • FastAPI • Streamlit

Purpose: Practical guide to the implemented database, financial analytics, screener, peers, valuation, clustering, reports, dashboard and API.

Status basis: Day 45 acceptance-gate results recorded during project verification, including documented limitations.

1. Project Overview

Nifty100 Financial Intelligence is an analytical platform built around a SQLite database, financial-ratio engine, screener, peer engine, valuation outputs, cash-flow intelligence, clustering, company tearsheets, Streamlit dashboard and FastAPI REST API.

Coverage

- 92 companies in the authoritative companies table.
- 84/92 companies (91.3%) have at least 10 years of P&L;, balance-sheet and cash-flow coverage.
- 1,188 financial-ratio rows.
- 11 peer groups with percentile data.
- 92 cluster assignments across cluster IDs 0–4.
- 483 generated Pros/Cons signals covering all 92 companies.
- 92 company tearsheet PDFs.
- 135 automated tests passed with zero failures at AC-18.

Technology

Layer	Implementation
Database	SQLite — nifty100.db
Analytics	Python, Pandas, NumPy, scikit-learn
Visualization	Matplotlib, Plotly
PDF	ReportLab
Dashboard	Streamlit
API	FastAPI + Uvicorn
Testing	pytest
Quality	Black + Ruff

2. Data Model and Data Quality

Core tables

- companies — authoritative company master and company-level metrics.
- profitandloss — historical P&L.;
- balancesheet — historical balance-sheet records.
- cashflow — historical cash-flow records.
- financial_ratios — calculated KPIs and ratios.
- sectors — implemented peer-group-based sector grouping.
- peer_groups — peer membership.
- peer_percentiles — peer-relative percentile metrics.
- market_cap — market-cap and valuation support data.
- valuation_summary — valuation outputs and flags.
- analysis, prosandcons and documents — supporting analytical/reporting data.

Quality controls

- Foreign-key validation returned zero errors at AC-03.
- 84/92 companies met the ≥ 10 -year coverage threshold.
- The ratio engine contains 1,188 rows.
- All 11 peer groups have percentile data.
- All 92 companies have cluster assignments.

Validation warning

The validation output records one WARNING under DQ-16 because one company has fewer than five years of ratio data. The warning is documented rather than silently removed.

3. Financial Analysis

Metric	Analyst use
ROE	Return relative to equity; used in quality/screener analysis.
Debt-to-Equity	Leverage indicator used in quality filtering.
Operating Profit Margin	Operating profitability.
Revenue CAGR	Multi-year revenue growth.
PAT CAGR	Multi-year profit growth.
EPS CAGR	Multi-year earnings-per-share growth.
Interest Coverage	Interest-obligation coverage indicator.
Asset Turnover	Asset-use efficiency.
Dividend Payout	Share of earnings distributed as dividends.

Manual validation

AC-05 manually validated TCS Revenue CAGR from Mar 2019 to Mar 2024 at 10.4636%, matching the database value within the required 0.1% tolerance.

Screener

The implemented screener supports minimum ROE, maximum debt-to-equity, minimum free cash flow, peer-group sector, minimum 5-year revenue CAGR, minimum 5-year PAT CAGR and maximum P/E. Dashboard results can be downloaded as UTF-8 CSV.

4. Screener and Peer Analysis

Quality screening

AC-07 returned 32 companies for ROE > 15, debt-to-equity < 1 and free cash flow > 0, within the required 10–50 range.

Peer group	Companies
Automobiles	7
Consumer Finance	3
FMCG	7
IT Services	5
Life Insurance	4
Oil & Gas	5
Pharmaceuticals	5
Power & Utilities	7
Private Banks	5
Public Sector Banks	4
Steel	4

Peer percentiles

AC-14 confirmed percentile data for all 11 peer groups. The peer_percentiles table supports relative KPI comparison within these groups.

5. Valuation and Cash-Flow Intelligence

Valuation flag	Count
Fair	48
Discount	30
Caution	14

Valuation fields

The valuation output contains company ID, company name, sector, P/E, P/B, EV/EBITDA, FCF yield, 5-year median P/E, P/E versus sector median and flag.

Cash-flow intelligence

- 92 authoritative companies covered.
- 1,152 deduplicated cash-flow rows.
- 62 CFO Quality High, 12 Moderate, 17 Accrual Risk, 1 unavailable.
- Capital intensity classifications are included in the intelligence output.

Capital allocation

Company tearsheets display the generated capital-allocation pattern; unavailable source values are shown as Unavailable.

6. Clustering and Company Archetypes

Clustering uses return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, fcf_cagr_5yr and operating_profit_margin_pct. Missing values are imputed, features are standardized, and KMeans uses five clusters with random_state=42.

- Elbow analysis is generated.
- cluster_labels.csv contains company_id, cluster_id, cluster_name and distance_from_centroid.
- All 92 companies have cluster assignments.

ID	Implemented cluster name
0	Value Cyclical
1	High-Quality Compounders
2	Distressed or Turnaround
3	Emerging Growth
4	Defensive Dividend Payers

Cluster names are analytical archetypes produced by the project workflow, not investment recommendations.

7. Company Tearsheets

Each tearsheet contains a company header, six KPI tiles, revenue/profit chart, ROE/ROCE chart, balance-sheet composition, cash-flow chart, Pros/Cons and capital-allocation pattern.

Coverage

AC-17 confirmed 92 PDF files in reports/tearsheets and no PDF below 30 KB.

JIOFIN

JIOFIN has only two proper historical P&L; periods after TTM is excluded. Its tearsheet uses available data without fabricating historical periods. The revenue/profit chart title is dynamic for limited-history companies.

Analyst note

Historical conclusions should be limited to the periods actually available for each company.

8. Streamlit Dashboard

Screen	Purpose
Home	Project overview and navigation.
Profile	Individual company profile and financial information.
Screener	Financial filters and CSV download.
Peers	Peer-group comparison.
Trends	Historical financial trends.
Sectors	Peer-group/sector analysis.
Capital	Cash-flow and capital-allocation intelligence.
Reports	Generated reports and outputs.

Suggested workflow

- Start at Home.
- Review a company in Profile.
- Use Screener for filtering.
- Use Peers for relative context.
- Use Trends for historical review.
- Use Capital for cash-flow intelligence.
- Open Reports for detailed PDFs.

9. REST API

Area	Route
Health	GET /api/v1/health
Companies	GET /api/v1/companies
Profile	GET /api/v1/companies/{ticker}
P&L	GET /api/v1/companies/{ticker}/profit-loss
Balance sheet	GET /api/v1/companies/{ticker}/balance-sheet
Cash flow	GET /api/v1/companies/{ticker}/cash-flow
Ratios	GET /api/v1/companies/{ticker}/ratios
Screener	GET /api/v1/screener
Sectors	GET /api/v1/sectors
Peers	GET /api/v1/peers
Company peers	GET /api/v1/peers/{ticker}
Valuation	GET /api/v1/valuation and /api/v1/valuation/{ticker}
Portfolio	GET /api/v1/portfolio
Documents	GET /api/v1/documents and /api/v1/documents/{ticker}

AC-08 measured the TCS profile at approximately 0.023 seconds. AC-11 returned HTTP 200 with database status ok. AC-12 returned 13 TCS ratio records.

10. Testing and Acceptance

AC-18 completed with 135 passed tests, zero failures and zero errors in 2.61 seconds. Two warnings were reported: a Starlette/AnyIO deprecation warning and a pytest cache warning caused by the Windows NUL configuration workaround.

Gate	Result	Evidence
AC-01	PASS	92 companies
AC-02	PASS	84/92 >=10-year coverage
AC-03	PASS	0 FK errors
AC-04	PASS	1,188 ratios
AC-05	PASS	TCS CAGR validated
AC-06	FAIL	ROE methodology/source difference
AC-07	PASS	32 quality-screen matches
AC-08	PASS	TCS profile <3 sec
AC-09	PASS	CSV download verified
AC-10	PASS	PDF checks completed
AC-11	PASS	Health HTTP 200
AC-12	PASS	13 TCS ratio records
AC-13	FAIL	API/Excel screener mismatch
AC-14	PASS	11 peer groups
AC-15	PASS	92 cluster assignments
AC-16	PASS	92 Pro + 92 Con
AC-17	PASS	92 PDFs >=30 KB
AC-18	PASS	135 tests, 0 failures
AC-19	PASS	Required CSV schema
AC-20	PASS	This guide >=10 pages

11. Known Limitations and Analyst Notes

AC-06

The acceptance comparison between `companies.roe_percentage` and `financial_ratios.return_on_equity_pct` did not consistently fall within the requested 5% difference. Investigation showed the ratio-engine value can be reproduced from underlying financial-statement fields for sampled companies, while the companies table contains a separate pre-existing ROE value. Both values were retained.

AC-13

The Quality-screen API result did not match the existing `screener_output.xlsx` company set. The API uses the implemented companies-level ROE together with latest ratio-engine leverage/free-cash-flow filters, while the Excel output came from an earlier screener workflow. This remains documented as a validation difference.

Analyst guidance

- Check period and data coverage before drawing trend conclusions.
- Distinguish company-level fields from ratio-engine calculations when comparing ROE.
- Use peer groups for context alongside underlying financial statements.
- Treat cluster names as analytical archetypes, not recommendations.
- Investigate discrepancies between API and older workbook outputs before relying on them interchangeably.

Status at guide creation

18 acceptance gates pass and 2 remain failed: AC-06 and AC-13. AC-20 is satisfied by this analyst guide once the generated PDF is verified for page count.